

8.6 Consider the wage equation

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + e_i \quad (XR8.6a)$$

where wage is measured in dollars per hour, education and experience are in years, and $METRO = 1$ if the person lives in a metropolitan area. We have $N = 1000$ observations from 2013.

- a. We are curious whether holding education, experience, and $METRO$ constant, there is the same amount of random variation in wages for males and females. Suppose $\text{var}(e_i | x_i, FEMALE = 0) = \sigma_M^2$ and $\text{var}(e_i | x_i, FEMALE = 1) = \sigma_F^2$. We specifically wish to test the null hypothesis $\sigma_M^2 = \sigma_F^2$ against $\sigma_M^2 \neq \sigma_F^2$. Using 577 observations on males, we obtain the sum of squared OLS residuals, $SSE_M = 97161.9174$. The regression using data on females yields $\hat{\sigma}_F^2 = 12.024$. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.
- b. We hypothesize that married individuals, relying on spousal support, can seek wider employment types and hence holding all else equal should have more variable wages. Suppose $\text{var}(e_i | x_i, MARRIED = 0) = \sigma_{SINGLE}^2$ and $\text{var}(e_i | x_i, MARRIED = 1) = \sigma_{MARRIED}^2$. Specify the null hypothesis $\sigma_{SINGLE}^2 = \sigma_{MARRIED}^2$ versus the alternative hypothesis $\sigma_{MARRIED}^2 > \sigma_{SINGLE}^2$. We add $FEMALE$ to the wage equation as an explanatory variable, so that

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + \beta_5 FEMALE + e_i \quad (XR8.6b)$$

Using $N = 400$ observations on single individuals, OLS estimation of (XR8.6b) yields a sum of squared residuals is 56231.0382. For the 600 married individuals, the sum of squared errors is 100,703.0471. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

- c. Following the regression in part (b), we carry out the NR^2 test using the right-hand-side variables in (XR8.6b) as candidates related to the heteroskedasticity. The value of this statistic is 59.03. What do we conclude about heteroskedasticity, at the 5% level? Does this provide evidence about the issue discussed in part (b), whether the error variation is different for married and unmarried individuals? Explain.
- d. Following the regression in part (b) we carry out the White test for heteroskedasticity. The value of the test statistic is 78.82. What are the degrees of freedom of the test statistic? What is the 5% critical value for the test? What do you conclude?

a. $N_M = 577$, $df_M = 577 - 4 = 573$, $df_F = 423 - 4 = 419$

$$H_0 : \sigma_M^2 = \sigma_F^2 , H_1 : \sigma_M^2 \neq \sigma_F^2$$

$$\hat{\sigma}_M^2 = \frac{SSE_M}{df_M} = \frac{97161.9174}{573} = 169.57 \quad \hat{\sigma}_F^2 = (12.024)^2 = 144.5766$$

$$GQ = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_F^2} = \frac{169.57}{144.5766} = 1.1729$$

$$F(573, 419, 0.025) = 0.8377$$

$$F(573, 419, 0.0975) = 1.1968$$

$$\Rightarrow \begin{aligned} & \because 0.8377 < 1.1729 < 1.1968 \\ & \therefore \text{do not reject } H_0 \end{aligned}$$

b. $N_{sin} = 400$, $N_{mar} = 600$

$$df_{sin} = 400 - 5 = 395 , df_{mar} = 600 - 5 = 595$$

$$SSE_{sin} = 56231.0382 , SSE_{mar} = 100703.0471$$

$$H_0 : \sigma_{mar}^2 = \sigma_{sin}^2 , H_1 : \sigma_{mar}^2 > \sigma_{sin}^2$$

$$\hat{\sigma}_{sin}^2 = \frac{56231.0382}{395} = 142.3571$$

$$\hat{\sigma}_{mar}^2 = \frac{100703.0471}{595} = 169.2488$$

$$GQ = \frac{\hat{\sigma}_{mar}^2}{\hat{\sigma}_{sin}^2} = \frac{169.2488}{142.3571} = 1.1889$$

$$F(595, 395, 0.95) = 1.1647 \quad \therefore 1.1889 > 1.1647 \quad \therefore \text{reject } H_0$$

c. H_0 : homoskedasticity H_1 : heteroskedasticity

$$df = 5 - 1 = 4$$

$$\chi^2_{(0.95, 4)} = 9.488 < NR^2 = 59.03$$

$\therefore$ reject the null hypothesis of homoskedasticity in the pooled regression.

d. 原始變數: EDUC, EXPER, METRO, FEMALE/MALE

連續變數平方項及交乘項: EDUC², EXPER², EDUC x EXPER

連續 x 虛擬: EDUC x MARRIED, EXPER x MARRIED

EDUC x FEMALE, EXPER x FEMALE

虛擬 x 虛擬: FEMALE x MARRIED

$$df = 4 + 3 + 4 + 1 = 12$$

$$\chi^2_{(12, 0.95)} = 21.0261 < 78.82$$

$\therefore$ reject H_0

reject the null hypothesis of homoskedasticity in the pooled regression.

e. The OLS fitted model from part (b), with usual and robust standard errors, is

$$\widehat{WAGE} = -17.77 + 2.50EDUC + 0.23EXPER + 3.23METRO - 4.20FEMALE$$

(se)	(2.36)	(0.14)	(0.031)	(1.05)	(0.81)
(robse)	(2.50)	(0.16)	(0.029)	(0.84)	(0.80)

For which coefficients have interval estimates gotten narrower? For which coefficients have interval estimates gotten wider? Is there an inconsistency in the results?

f. If we add MARRIED to the model in part (b), we find that its t -value using a White heteroskedasticity robust standard error is about 1.0. Does this conflict with, or is it compatible with, the result in (b) concerning heteroskedasticity? Explain.

e. 截距項和EDUC1系數的區間估計變得更寬

其他變數的區間估計則變得更窄

並不予盾, robust se 有更能以錯誤的OLS標準誤來得更大, 或更小

f. 用MARRIED 檢驗是否對薪資的平均影響顯著, $t=1$ ∴ 不顯著

但(b)是用F test 檢驗已婚者2組人在薪資的變動程度上不同

是為異質變異數的來源, 2者為不同概念, 並不予盾

8.16 A sample of 200 Chicago households was taken to investigate how far American households tend to travel when they take a vacation. Consider the model

$$MILES = \beta_1 + \beta_2 INCOME + \beta_3 AGE + \beta_4 KIDS + e$$

MILES is miles driven per year, *INCOME* is measured in \$1000 units, *AGE* is the average age of the adult members of the household, and *KIDS* is the number of children.

- Use the data file *vacation* to estimate the model by OLS. Construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant.
- Plot the OLS residuals versus *INCOME* and *AGE*. Do you observe any patterns suggesting that heteroskedasticity is present?
- Sort the data according to increasing magnitude of income. Estimate the model using the first 90 observations and again using the last 90 observations. Carry out the Goldfeld-Quandt test for heteroskedastic errors at the 5% level. State the null and alternative hypotheses.
- Estimate the model by OLS using heteroskedasticity robust standard errors. Construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant. How does this interval estimate compare to the one in (a)?
- Obtain GLS estimates assuming $\sigma_i^2 = \sigma^2 INCOME_i^2$. Using both conventional GLS and robust GLS standard errors, construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant. How do these interval estimates compare to the ones in (a) and (d)?

a.

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Call:
lm(formula = miles ~ income + age + kids, data = vacation)

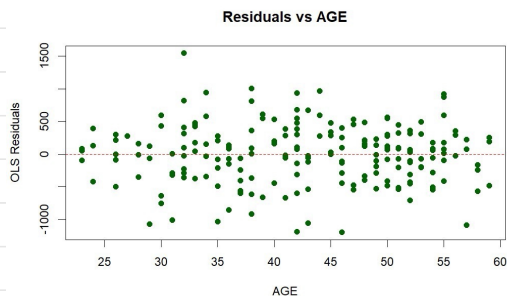
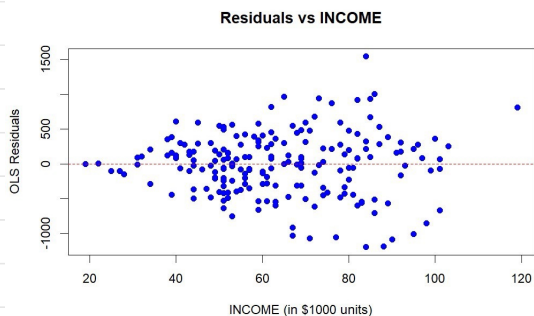
Residuals:
    Min       1Q   Median       3Q      Max
-1198.14  -295.31   17.98   287.54  1549.41

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -391.548    169.775   -2.306  0.0221 *
income       14.201      1.800    7.889 2.10e-13 ***
age          15.741      3.757    4.189 4.23e-05 ***
kids        -81.826     27.130   -3.016  0.0029 **
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 452.3 on 196 degrees of freedom
Multiple R-squared:  0.3406,    Adjusted R-squared:  0.3305
F-statistic: 33.75 on 3 and 196 DF,  p-value: < 2.2e-16

      2.5 %    97.5 %
kids -135.3298 -28.32302
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b. 隨收入增加，殘差的變異增加。殘差對 age 則不明顯



c. --- Goldfeld-Quandt Test Results ---
 RSS1 (較低收入組): 8750040
 RSS2 (較高收入組): 27160654
 F-statistic: 3.104061
 Degrees of Freedom: 86 and 86
 p-value: 1.64001e-07

結論：在 5% 顯著水準下拒絕 H_0 ，有證據顯示存在異質變異數。有充份的統計證據支持殘差的變異數隨收入增加而增大。

d. t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-391.5480	142.6548	-2.7447	0.0066190 **
income	14.2013	1.9389	7.3246	6.083e-12 ***
age	15.7409	3.9657	3.9692	0.0001011 ***
kids	-81.8264	29.1544	-2.8067	0.0055112 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

--- 95% Robust Interval Estimate for KIDS ---
 區間估計: [-139.3230, -24.3299]

和 a. 相比，具較大的 se，所以區間更寬

e.

Call: lm(formula = miles ~ income + age + kids, data = vacation, weights = 1/(income^2))

Weighted Residuals:

	Min	1Q	Median	3Q	Max
	-15.1907	-4.9555	0.2488	4.3832	18.5462

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-424.996	121.444	-3.500	0.000577 ***
income	13.947	1.481	9.420	< 2e-16 ***
age	16.717	3.025	5.527	1.03e-07 ***
kids	-76.806	21.848	-3.515	0.000545 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.765 on 196 degrees of freedom
 Multiple R-squared: 0.4573, Adjusted R-squared: 0.449
 F-statistic: 55.06 on 3 and 196 DF, p-value: < 2.2e-16

--- 傳統 GLS 95% 信賴區間 ---

	2.5 %	97.5 %
	-119.89450	-33.71808

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-424.9962	95.8035	-4.4361	1.526e-05 ***
income	13.9473	1.3470	10.3545	< 2.2e-16 ***
age	16.7175	2.7974	5.9761	1.061e-08 ***
kids	-76.8063	22.6186	-3.3957	0.0008286 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

--- Robust GLS 95% 信賴區間 ---
 區間估計: [-121.4134, -32.1992]



和 OLS 相比，GLS se 更小，區間更窄